The Essence of Gravity-The Origin of Mass-The role of SU(3) QCD and Higgs Mechanism

A speculative framework

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ABSTRACT: Our previous research ("The Arrow of Time under Time and Entropy Mapping Model") established a discrete spacetime model mapping multiplicative entropy to time evolution, providing an entropy coordinate for computational physics simulations. This study develops the spacetime quantization framework based on Space Elementary Quanta (SEQ), proposing that gravitational interactions fundamentally arise from translational deformations of the SEQ network. The model reveals that mass originates as stored gravitational potential energy through SU(3)-mediated spatial compression, while gravity emerges as consequential stretching of surrounding SEQ structure - providing a unified mechanism for both inertial and gravitational mass. Building on general relativity's curvature framework but from a discrete spacetime perspective, we demonstrate how equivalent gravitational fields generated by acceleration share this same origin through directional SEQ compression. The theory further offers a concrete physical interpretation of SU(3) color interactions as dynamic spatial compression modes within hadrons, establishing a direct connection between quantum chromodynamics and gravitational phenomena.The Higgs field acts as a quantum chiral lock, stabilizing the SU(3)-mediated elastic energy storage in localized space by enforcing symmetry-breaking constraints. Through this approach, we construct a tentative unification framework where both general relativity's spacetime curvature and quantum field theory's gauge interactions emerge from the underlying dynamics of quantized space.

Key words:the Essence of Gravitation;the Origin of Mass;the role of SU(3) Color Interaction;Higgs chiral lock;

I. INTRODUCTION.

This work proposes a novel unification framework that bridges the chasm between quantum field theory and general relativity by revealing their shared geometric foundation in spacetime deformation dynamics. While the Standard Model successfully unifies electromagnetic, weak, and strong interactions through $U(1)\times SU(2)\times SU(3)$ gauge symmetries, its integration with gravity has remained elusive. Here we demonstrate that both color confinement in QCD and gravitational curvature fundamentally arise from compression and stretching of space elementary quanta (SEQ) network- with SU(3) interactions generating mass through local spatial compression and gravity emerging via non-local SEQ network deformation. The Higgs field stabilizes this compression via symmetry breaking, acting as a chiral "quantum lock" to prevent energy dissipation. This geometric correspondence suggests a deeper connection between gauge theory and gravitation than previously recognized, where hadronic matter curves spacetime through the same mechanism that creates its mass-energy content. Although presented at a conceptual level, the framework's consistency with established phenomena (including asymptotic freedom, gravitational lensing, and the equivalence principle) provides compelling motivation for future mathematical formalization. The proposed SEQ-mediated unification maintains compatibility with existing quantum and relativistic frameworks, potentially opening new pathways toward quantum gravity.

II. Derivation of the Color Interaction-Mass-Gravity Relationship

Building upon the geometric foundation of general relativity, where gravity emerges from spacetime curvature, we propose a fundamental connection between the origin of mass, gravitational effects, and the SU(3) color interaction in the Standard Model. The logical derivation proceeds as follows:

(i)General relativity establishes that gravitational fields arise from the curvature of spacetime, which is sourced by mass-energy. This implies that massive objects must inherently distort local spacetime geometry.

(ii)If gravitational effects manifest as the stretching of spacetime in regions surrounding massive bodies, then mass itself must correspond to a localized compression of the underlying quantum spacetime structure — described in our framework as the compression of SEQ network.

(iii)Within hadrons, the dominant interaction governing quark dynamics is the SU(3) color force. We posit that this interaction fundamentally mediates the high-density compression of SEQ, generating both:

-Mass: The stored elastic potential energy of the compressed SEQ network.

-Gravity: The consequential stretching of surrounding spacetime.

(iv)The structure of SU(3) ensures that this compression maintains approximate spherical symmetry.

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(V)The framework naturally extends to : acceleration-induced "equivalent" gravitational fields arise from directional SEQ compression along the acceleration axis, consistent with general relativity but with explicit microscopic dynamics.

III. Fundamental Postulates

3.1 Cosmological Expansion
The universe exhibits continuous expansion.

3.2 Thermodynamic Constraints
The universe's evolution is governed by the law of energy conservation and the law of entropy increase.

3.3 Spacetime Quantum Structure
The universe consists of space elementary quanta (SEQ) forming a 3D topological manifold where adjacency relations are preserved despite local energy variations. Gravitational lensing around black holes reveals coupling between gravitational and electromagnetic field quanta, suggesting they originate from different excitation states of a unified quantum field - the SEQ hypothesis. The equilibrium spacing and tension between adjacent SEQ can be dynamically modified by gravitational or equivalent fields.

3.4 Field-Particle Correspondence
All field quanta and elementary particles emerge as distinct excitation modes of SEQ, mathematically describable as 3D dynamical structural matrices of SEQ.

3.5 Ground State Properties
Each SEQ possesses a non-zero ground state energy (e.g., spin or vibrational modes). Fixed ground-state spin chirality may explain parity violation in weak interactions, while the collective ground-state energy could provide a microscopic origin for the cosmological constant in Einstein's field equations. This framework shows conceptual overlap with Loop Quantum Gravity.

3.6 Spacetime Elasticity
Neighboring SEQ maintain metastable equilibrium separation. Deviations alter the characteristic resonant frequency of SEQ and generate restoring forces (manifesting as gravity/equivalent gravity). The nonlinear, asymmetric response function can produce spatially varying gradients, resembling Einstein's elastic continuum description and Zee's discrete spring network analogy.

3.7 Nature of Motion
Since black holes preserve local spacetime connectivity despite extreme curvature, apparent particle trajectories must represent propagating spacetime excitations rather than topological rearrangement. This interpretation remains consistent with General Relativity, with the speed of light (c) emerging as the upper bound for excitation propagation.

3.8 Adjacent SEQs maintain a dynamic equilibrium spacing interconnected via spring-like bonds in their ground state. SEQs are fundamental, indivisible Planck-scale entities—their structure remains intact under any deformation or energy fluctuations.
The sub-Planckian regime governs the spatial elasticity of the network (including elastic potential storage and release), while quantized energy transfer occurs exclusively through interactions between SEQs. Within this framework:

- The spin degrees of freedom of SEQs and their elastic bonds remain decoupled, preserving independent dynamical regimes.
- Under perturbation, the system responds by modifying SEQ resonant frequencies while generating compressive/tensile forces.
- This response is nonlinear and asymmetric, enabling emergent behaviors (e.g., regional gradient variations).
- SEQs are stable, indivisible structures composed of sub-Planckian components. SEQs' spin emerges from collective space transformations at the sub-Planck level. This ensures the spin degrees of freedom do not interfere with elastic deformations in the SEQ network. This architecture naturally protects spin dynamics from elastic disturbances.
- At the sub-Planckian scale, the elastic properties of the underlying substrate impose an upper bound on the spacing modulation and tension between adjacent SEQs. This fundamental limit ensures that extreme deformations (e.g., near black hole singularities) cannot disrupt the topological integrity of the SEQ network.

IV.Gravitational Interaction and general relativity

4.1 Gravitational Interaction
We propose that gravitational effects fundamentally arise from discrete translational deformations of SEQ network, where the interaction manifests through modifications of equilibrium SEQ spacing. This mechanism operates through several key features: (i) The SEQ network exhibits spring-like bonding in its ground state, with gravitational potentials corresponding to stored elastic energy from inter-SEQ tension; (ii) Local spacetime curvature

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emerges when this tension modifies the harmonic oscillation frequencies of SEQs, causing both the observed gravitational redshift (through reduced oscillation frequencies) and time dilation effects that exactly match general relativistic predictions. The interaction propagates via spherically divergent translational strain waves, with strain density following an inverse-square law due to geometric dispersion over expanding wavefront surfaces - naturally reproducing Newtonian gravity at macroscopic scales. When operating on initially flat spacetime, this action induces effective curvature through topologically homeomorphic transformations of the SEQ lattice structure, generating metric perturbations identical to those described by Einstein's field equations. Crucially, the model introduces an intrinsic upper limit to gravitational potential energy through nonlinear SEQ bonding elasticity, which (i) prevents singularities at astrophysical scales, (ii) modifies black hole thermodynamics near event horizons, and (iii) suggests testable deviations from classical GR in extreme curvature regimes. The framework maintains full compatibility with established gravitational phenomena while providing microscopic explanations for both the equivalence principle (through universal SEQ strain-response dynamics) and the holographic principle (via discrete surface-to-volume scaling of SEQ interactions).

4.2 Newton's Law of Universal Gravitation

The gravitational interaction strength is fundamentally determined by the surface density of space elementary quanta (SEQs) associated with a mass source. Each surface SEQ establishes a discrete gravitational conduction pathway, corresponding microscopically to a line of gravitational force. These pathways exhibit spherical divergence from the source, with their areal density decaying as r^{-2} — a geometric relationship that naturally recovers the inverse-square dependence in Newton's law of universal gravitation.

4.3 **Table 1: Correspondence with GR**

General Relativity Concept	SEQ Framework Interpretation	Key Features
Metric Field	Dynamic deformation of 3D SEQ structural matrix and local density variation	- Describes gravitational lensing - Explains dark matter gravitational effects
Geodesic Path	Minimum cumulative conduction count path through distorted SEQ network	- Satisfies least action principle - Accounts for light bending near massive objects
Proper Time Dilation	Reduced local SEQ transformation frequency under gravitational/kinematic stress	- Matches GR time dilation predictions - Mechanism: decreased oscillation frequency in SEQ spacing variation
Cosmological Constant	Macroscopic manifestation of SEQ ground state energy	- Relates to dark energy - Explains parity violation through fixed spin chirality
Diffeomorphism Invariance	Global topological homeomorphic transformation of SEQ network	- Maintains adjacency relations - Analogous to continuum fluid mechanics
Event Horizon	SEQ spacing compressed to elastic limit (maximum tension)	- Halts local transformations - Naturally prevents singularities
Equivalence Principle	Directional SEQ compression matching acceleration/gravitational effects	- Unified mechanism for inertial and gravitational mass

V.Understanding on standard model of quantum field theory and link between gravitation, equivalent gravitation and SU(3)

Deriving the Duality: How SU(3) Compression Generates Mass While Stretching Spacetime Produces Gravity.

General Relativity establishes gravitational fields as spacetime curvature →This curvature originates from localized metric perturbations →Mass must therefore represent a concentrated deformation of spacetime itself → creating self-consistent stress-energy sourcing of curvature → Within hadronic matter, this deformation is actively mediated by SU(3) color interactions → The SU(3) force compresses SEQ network to extreme densities → This compression stores energy as hadronic mass while simultaneously → inducing outward stretching of surrounding SEQ

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networks → manifested macroscopically as gravitational attraction→ Crucially, SU(3) enforces spherical symmetry during compression → which naturally produces the observed isotropic gravitational field → This mechanism extends to equivalent cases: relative motion anisotropically compresses local SEQ arrangements → generating acceleration-induced "gravitational" effects through adjacent SEQ tension gradients.

5.1 SU(3) Phase: Corresponds to compressive /stretching deformations of the inter-SEQ spacings.

5.2 U(1) Dynamics: Emerges from rotational excitations governing electromagnetic interactions.

5.3 SU(2) Mechanism: Represents adjustments of rotational axes and spin in charged particle structural matrices, with their inherent chirality encoding weak interaction symmetry breaking. The differential coupling between these chiral matrices and the fixed chirality of SEQ ground-state spins underlies the observed parity violation. While SU(2) lacks explicit chirality parameters, its definition through axis rotations and spin of chirally-specific charged particles intrinsically incorporates these degrees of freedom.

5.4 SU(3):

5.4.1 Quark Structure Dynamics

The three-dimensional quark matrix structure is characterized by a multi-layered, multi-axial rotational configuration. The high energy density within this structure maintains the Space Elementary Quanta (SEQs) in a dynamic equilibrium between compression and stretching states, with interlayer interactions similarly preserving this equilibrium.

5.4.2 Fractional Charge and Layered Architecture

The fractional charges of quarks originate from their structural stratification:

(i)Quarks with +2/3 charge possess twice the number of structural layers compared to -1/3 charged quarks

These configurations exhibit distinct chiral properties in their matrix representations

(ii)Proton and neutron charge quantization emerges from specific stratified arrangements where the inter-quark layer spacing exceeds intra-quark layer distances.

5.4.3 Color Charge and Hadronic Structure

The color property manifests as:The principal axis of maximum energy density in quark structural matrices.Hadronic color neutrality corresponds to global spatial symmetry and electric field isotropy.

Antiquarks are represented by parity-inverted rotational transformations of quark matrices.

5.4.4 Gluon-Mediated Interactions

The eight generators of SU(3) correspond to distinct interaction types: (i)Six off-diagonal generators mediate color exchange operations combined with compression/stretching phase transformations.

(ii)Two diagonal generators govern scaling transformations across color dimensions.

(iii)These interactions primarily occur within the interlayer regions of hadronic structures.

5.4.5 Mass Generation Mechanism

The color dynamics described by SU(3) generators - particularly through three-dimensional color distribution modulation - demonstrates fundamental coupling to gravitational effects. This suggests that the mechanism of color charge exchange and modulation may contribute to the origin of mass.

5.4.6 Gluons serve as mediators of both compressive and tensile stresses between quarks and interlayer SEQs. These gauge bosons can be conceptualized as quasi-structures composed of highly condensed SEQs, analogous to a dense array of dynamical springs.

5.4.7 The observed phenomena of quark asymptotic freedom and color confinement emerge fundamentally from nonlinear stress dynamics in the SEQ medium.

5.4.8 The eight SU(3) generators act as spacetime deformation operators that perform three key functions simultaneously: (1) compress local space through non-Abelian gauge operations, (2) maintain approximate spherical symmetry via multidimensional stress modulation, and (3) induce gravitational effects through spacetime strain. This dynamic compression-stretching mechanism explains why mass always generates isotropic gravity - the SU(3) operations actively enforce spherical symmetry during space compression, causing the resulting external spacetime stretching (gravity) to inherit this symmetry.

5.4.9 The three-quark point-like configuration inherently fails to achieve spatial symmetry, contradicting the observed spherical charge distribution of protons, whereas this hypothesis of a layered arrangement in a quasi-spherical structure of up and down quarks within the proton offers a more plausible explanation for the integer charge of the proton and the isotropic nature of the electric field as well.

5.5 The essence of mass:

The stored elastic(gravitational) potential energy of the compressed SEQ network under SU(3) symmetry.

5.5.1 Dimensional analysis dictates that the relationship between mass and energy must satisfy [E]/[m][v^2], with the proportionality coefficient determined by the fundamental constants of spacetime (the speed of light, c).

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5.5.2 The compressed potential energy of mass in localized space is inherently released as mainly gravitational waves with radiation, which propagate at speed c , thus directly yielding $\Delta E=\Delta mc^2$.

5.6 The SU(3) color interaction and gravitational field generation exhibit a fundamental duality through spacetime quantization: The mass-energy of hadrons emerges from non-Abelian gauge-driven compaction, while gravity arises from the resultant strain field propagating through the finite quantum spacetime fabric. This intrinsic separation between localized color confinement and global SEQ network deformation underpins the hierarchical strength disparity of the two interactions.

5.7 Complementary **Role of the Higgs Field**: Symmetry Breaking and "Locking" Mechanism:

5.7.1 The Higgs field plays a crucial yet subtle role in this framework by acting as a stabilizing "quantum chiral lock" that preserves the compression effects mediated by the SU(3) gluon field on the Space Elementary Quanta (SEQ). While the SU(3) color force actively compresses the SEQ network to generate mass-energy through spatial deformation, the Higgs mechanism serves to maintain this compressed configuration in a stable equilibrium state. This locking function is particularly vital for quark confinement, as it prevents the rapid dissipation of the gluon field's compressive energy that would otherwise lead to deconfinement. The Higgs field's symmetry-breaking properties thus complement the SU(3) compression mechanism by providing an additional layer of stability to the mass-generating structure. In essence, if the SU(3) mediated compression is likened to a tensed spring storing potential energy, the Higgs field acts as the catch mechanism that keeps the spring compressed, ensuring the persistence of the mass effect. This dual mechanism - active compression by color forces and passive stabilization by the Higgs field - offers a more complete picture of mass generation that bridges quantum chromodynamics with electroweak theory while remaining consistent with the discrete spacetime framework proposed in the paper. The interplay between these mechanisms may also help explain why certain particles (like quarks) exhibit both confinement and mass properties, while others (like leptons) primarily acquire mass through Higgs interactions alone.

5.7.2 This dual mechanism—where the QCD color interaction-SU(3) acts as a compressive spring system, while the Higgs mechanism functions like a preloaded torsional spring combined with a ratchet (enabling unidirectional energy storage while preventing reversal) - provides a vivid mechanical analogy for how fundamental particles maintain their mass stability in the quantum spacetime fabric. Just as a ratchet's teeth enforce unidirectional motion through asymmetric geometry, the Higgs' chiral coupling to the SEQ ground state may similarly lock the gluon field's compressive energy in an irreversible configuration.

VI Discussion

The proposed SEQ framework, while offering a unified mechanism for mass generation, gravitational effects, and SU(3) color interactions through spacetime quantization, remains fundamentally conceptual at this stage. Its advancement hinges on resolving two pivotal challenges: (1) establishing a discrete counterpart to Einstein's field equations that reproduces the SEQ network's elastodynamic behavior in the continuum limit, and (2) constructing a scale-dependent "space elasticity coefficient" function informed by both QCD-scale phenomena (e.g., quark confinement energy gradients) and cosmological observations (e.g., dark energy distribution anisotropies). These tasks likely exceed the capacity of independent research, requiring concerted effort from the broader physics community — particularly specialists in quantum gravity (to reconcile holographic principles with SEQ dynamics), particle phenomenology (to test the predicted chiral effects on lepton magnetic moments), and numerical relativity (to develop discrete spacetime simulations).

The path forward parallels theoretical breakthroughs where conceptual clarity preceded mathematical rigor: initial focus should center on disseminating the framework's core postulates to stimulate debate, followed by collaborative formalization of its equations, and ultimately observational tests of its novel predictions. The framework's value lies not in immediate completeness but in its capacity to reframe persistent problems, the quantum origin of mass, and the geometrization of gauge interactions—as solvable challenges within a shared spacetime microstructure paradigm. Crucially, it transforms abstract notions of "spacetime elasticity" into a concrete quantum network , inviting both theoretical refinement and experimental engagement.

VII Statement

It should be noted that the interpretation of mass-gravity-color dynamics presented here is essentially a reformulation of **Einstein's classical elastic spacetime** theory.

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The present discrete framework constitutes a complementary rather than competing paradigm to continuum-based gravitational theories. This pluralistic view recognizes that both approaches confront fundamental epistemological boundaries: (i) Discrete formulations must address the nature of inter-quantum intervals - whether these represent emergent geometric relationships or require deeper ontological commitments to a background structure. (ii) Continuum descriptions grapple with the operational indefinability of infinitesimals at Planck scales, where measurement theory itself breaks down.

The SEQ approach navigates this dichotomy by treating discreteness as a computational primitive while preserving topological homeomorphism - thus maintaining compatibility with general relativity's phenomenological successes.

This work extends the explanatory reach of the Standard Model by revealing a unified mechanism for quantum chromodynamics and gravity through spacetime quantization. The SU(3)-gravitational duality—where color confinement and emergent gravity originate from localized compression and global strain of quantum spacetime—preserves all Standard Model successes while incorporating gravitation.

The elastic spacetime paradigm, pioneered by Einstein's geometric intuition (Einstein, 1916)[1] has been profoundly developed by subsequent physicists through both theoretical refinements and experimental verification. My SEQ model humbly stands on these giants' shoulders by reinterpreting their continuum-based insights through a discrete quantum lens.

As is widely known, Einstein established General Relativity; Planck's discoveries ignited the discussion and development of quantum theory; and Dirac laid the foundational framework for Quantum Field Theory. Throughout this process, numerous esteemed physicists have developed and refined these frameworks. I am merely an amateur physics enthusiast and cannot individually enumerate all these great scientists and their works, I hereby pay my respects to them. This text represents my conceptual framework for understanding these issues. If this understanding gains recognition, its further development will rely on the efforts of professional physicists.

Murray Gell-Mann; George Zweig; Harald Fritzsch; Heinrich Leutwyler; David Gross[2]; Frank Wilczek[3]; Hugh Politzer[4] historically incorporated SU(3) into strong interactions, These physicists studying the SU(3) strong interaction not only integrated algebraic structures with complex physical phenomena but also uncovered intricate phenomena such as quark confinement and asymptotic freedom. Their work has significantly advanced the understanding of the fundamental forces governing particle interactions. Anthony Zee[5] actually proposed the physical intuition of spring networks decades ago in his renowned textbook Quantum Field Theory in a Nutshell (2003). Gerard't Hooft[7] has made groundbreaking advances in quantum gravity, advocating for deterministic, discrete structures underlying quantum mechanics and general relativity. Carlo Rovelli [8]and Lee Smolin[9] have made significant contributions through their development of loop quantum gravity In the realm of quantum gravity. Rafael Sorkin[10] is renowned for his foundational work on causal set theory, which posits that spacetime is fundamentally discrete and described by a partially ordered set of events; Edward Witten[11] has made transformative contributions to string theory and quantum field theory. Michael Turner[12] is celebrated for his pioneering research in cosmology, especially his work on dark energy. Xiao-Gang Wen[13] has made pioneering work on topological order and string-net condensation demonstrates. John Wheeler's[14] visionary concept of quantum foam—a fluctuating discrete spacetime at the Planck scale. Fotini Markopoulou's[15] Quantum Graphity model, in which spacetime emerges as a dynamical graph of quantum connections, parallels the SEQ framework. David Finkelstein,[16] a pioneer in discrete spacetime physics, proposed that time and space emerge from algebraic operations on fundamental quantum units. Wayne C. Myrvold's[17] seminal work to the philosophical foundations of thermodynamics have significantly deepened our understanding of its role as a fundamental driver of the universe. Recently, I noticed that Perez Felipe Sergio's[18] might have earlier recognized how the compression of space by massive objects leads to external stretching. Ali H. Chamseddine,Viatcheslav Mukhanov[19] develop a discrete differential geometry framework, where spacetime curvature and connections emerge from elementary Planck-scale cells, bridging discrete and continuous spacetime. Dmitry Chelkak, Alexander Glazman, Stanislav Smirnov [20]developed a discrete version of the stress-energy tensor for lattice models, rigorously connecting it to continuum field theory. I recently discovered through search engines that Gudrun Kalmbach H.E. (2021)[23] and C. G. Sim (2021)[24] reported the relationship between gravity and color charge interactions in their respective papers before

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me, although our theoretical frameworks differ fundamentally.

Similar ideas may exist in earlier literature. I strive to properly cite the relevant prior work as I know them through ongoing research.

VIII Reference:

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X Appendix

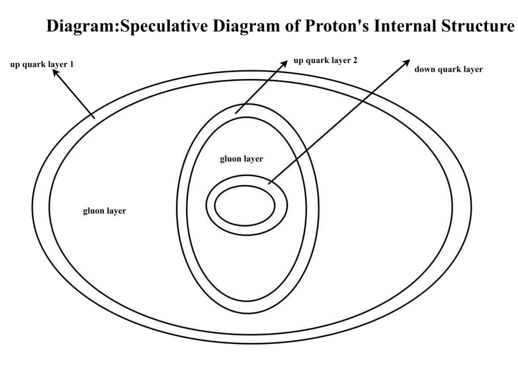

Diagram.1:Speculative Diagram of Proton's Internal Structure with Quarks and Gluons. Online Draw tool[21]

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